Main difference

Concept	`for` loop	`lambda`
What is it?	A **loop**	A **function**
Main purpose	**Repeat** an operation	**Define** a small operation
Takes items one by one?	✅ Yes	❌ No
Performs calculation?	Can	Defines the calculation
Repetition	✅	❌ By itself
Example	`for x in numbers:`	`lambda x: x * 2`
In our example	Gets `2`, `4`, `6`, `8` one by one	Defines `x × 2`

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Most important

FOR LOOP
↓
Takes items one by one
↓
2 → 4 → 6 → 8

LAMBDA
↓
Defines the operation
↓
x → x × 2

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Using `for` loop

numbers = [2, 4, 6, 8]

for x in numbers:
    print(x * 2)

Output:

4
8
12
16

Here the `for` loop takes each number one by one:

x = 2 → 2 × 2 = 4
x = 4 → 4 × 2 = 8
x = 6 → 6 × 2 = 12
x = 8 → 8 × 2 = 16

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Using `lambda`

First, create a lambda function:

double = lambda x: x * 2

This means:

> **Take `x` and multiply it by 2.**

Then we can use the same `for` loop:

numbers = [2, 4, 6, 8]

double = lambda x: x * 2

for x in numbers:
    print(double(x))

Output:

4
8
12
16

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Term	Definition
For loop	**Repeats a process for each item in a collection.**
Lambda function	**A small, one-line function that defines an operation.**
Iteration	**One complete execution/repetition of a loop.**

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So in your example:

- `for` **= repetition** 🔄
- `lambda` **= operation/function** ⚙️
- **iteration = each individual repetition** 1️⃣2️⃣3️⃣4️⃣

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